

1. CQs about the taxonomy of road signs

Competency question: What are the main categories of road signs in the Vienna Convention?

Method: DL Query

Protégé option: Direct subclasses

Translation:

RoadSign

Competency question: Is a danger warning sign a type of road sign?

Method: DL Query

Protégé option: Superclasses

Translation:

DangerWarningSign

Competency question: Is a prohibitory sign a type of regulatory sign?

Method: DL Query

Protégé option: Superclasses

Translation:

ProhibitorySign

Competency question: Is a mandatory sign a type of regulatory sign?

Method: DL Query

Protégé option: Superclasses

Translation:

MandatorySign

Competency question: Which classes are subclasses of RoadSign?

Method: DL Query

Protégé option: Subclasses

Translation:

RoadSign

Competency question: Which signs belong to the category of additional panels?

Method: DL Query

Protégé option: Subclasses

Translation:

`AdditionalPanel`

Competency question: Which signs are informative rather than regulatory?

Method: DL Query

Protégé option: Subclasses

Translation:

`InformativeSign`

2. CQs about road-sign properties

Competency question: What shapes are associated with each traffic sign category?

Method: DL Query + Python recommended

Protégé option: Subclasses

Translation:

`RoadSign and (hasShape some SignShape)`

Competency question: What is the numerical speed limit value represented by a speed limit sign?

Method: DL Query

Protégé option: Subclasses

Translation:

`RoadSign and (hasMaximumSpeedValue some xsd:decimal)`

Competency question: What is the minimum speed value represented by a compulsory minimum speed sign?

Method: DL Query

Protégé option: Subclasses

Translation:

`RoadSign and (hasMinimumSpeedValue some xsd:decimal)`

Competency question: What is the advisory speed value represented by an advisory speed sign?

Method: DL Query

Protégé option: Subclasses

Translation:

`RoadSign and (hasAdvisorySpeedValue some xsd:decimal)`

Competency question: What is the maximum permitted height represented by a height restriction sign?

Method: DL Query

Protégé option: Subclasses

Translation:

`RoadSign and (hasMaximumHeightValue some xsd:decimal)`

Competency question: What is the maximum permitted width represented by a width restriction sign?

Method: DL Query

Protégé option: Subclasses

Translation:

`RoadSign and (hasMaximumWidthValue some xsd:decimal)`

Competency question: What is the maximum permitted length represented by a length restriction sign?

Method: DL Query

Protégé option: Subclasses

Translation:

`RoadSign and (hasMaximumLengthValue some xsd:decimal)`

Competency question: What is the maximum permitted laden mass represented by a weight restriction sign?

Method: DL Query

Protégé option: Subclasses

Translation:

`RoadSign and (hasMaximumWeightValue some xsd:decimal)`

Competency question: What is the maximum permitted axle mass represented by an axle-weight restriction sign?

Method: DL Query

Protégé option: Subclasses

Translation:

`RoadSign and (hasMaximumAxleWeightValue some xsd:decimal)`

Competency question: What distance value is displayed on an additional panel?

Method: DL Query

Protégé option: Subclasses

Translation:

RoadSign and (hasDistanceValue some xsd:decimal)

Competency question: What textual information is displayed on an information sign?

Method: DL Query

Protégé option: Subclasses

Translation:

RoadSign and (hasTextualInformation some xsd:string)

Competency question: What destination text is displayed on a direction sign?

Method: DL Query

Protégé option: Subclasses

Translation:

RoadSign and (hasDestinationText some xsd:string)

Competency question: What service name is displayed on a service sign?

Method: DL Query

Protégé option: Subclasses

Translation:

RoadSign and (hasServiceName some xsd:string)

Competency question: What road-sign code is associated with a sign class?

Method: SPARQL

Translation:

```
SELECT ?property ?domain ?range
WHERE {
  ?property rdf:type owl:DatatypeProperty .
  ?property rdfs:domain ?domain .
  ?property rdfs:range ?range .
  FILTER(CONTAINS(STR(?property), "hasSignCode"))
}
```

Competency question: Which properties are functional?

Method: SPARQL

Translation:

```
SELECT ?property
WHERE {
  ?property rdf:type owl:FunctionalProperty .
}
ORDER BY ?property
```

Competency question: Which object properties are inverse properties of each other?

Method: SPARQL

Translation:

```
SELECT ?property ?inverseProperty
WHERE {
  ?property owl:inverseOf ?inverseProperty .
}
ORDER BY ?property
```

3. CQs about the pragmatics of road signs

Competency question: Which type of road user does a sign apply to?

Method: DL Query

Protégé option: Subclasses

Translation:

```
RoadSign and (appliesToRoadUser some RoadUser)
```

Competency question: Which type of vehicle does a regulatory sign apply to?

Method: DL Query

Protégé option: Subclasses

Translation:

```
RegulatorySign and (appliesToVehicle some Vehicle)
```

Competency question: What regulation is expressed by a regulatory sign?

Method: DL Query

Protégé option: Subclasses

Translation:

RoadSign and (expressesRegulation some TrafficRegulation)

Competency question: What prohibition is expressed by a prohibitory sign?

Method: DL Query

Protégé option: Subclasses

Translation:

RoadSign and (expressesProhibition some Prohibition)

Competency question: What obligation is expressed by a mandatory sign?

Method: DL Query

Protégé option: Subclasses

Translation:

RoadSign and (expressesObligation some Obligation)

Competency question: What restriction is expressed by a restriction sign?

Method: DL Query

Protégé option: Subclasses

Translation:

RoadSign and (expressesRestriction some Restriction)

Competency question: What hazard is indicated by a warning sign?

Method: DL Query

Protégé option: Subclasses

Translation:

RoadSign and (warnsAbout some RoadHazard)

Competency question: What road condition is indicated by a warning sign?

Method: DL Query

Protégé option: Subclasses

Translation:

RoadSign and (indicatesRoadCondition some RoadCondition)

Competency question: What priority rule is expressed by a priority sign?

Method: DL Query

Protégé option: Subclasses

Translation:

`RoadSign and (indicatesPriorityRule some PriorityRule)`

Competency question: What direction is indicated by a direction sign?

Method: DL Query

Protégé option: Subclasses

Translation:

`RoadSign and (indicatesDirection some Direction)`

Competency question: What destination is indicated by a direction sign?

Method: DL Query

Protégé option: Subclasses

Translation:

`RoadSign and (indicatesDestination some Destination)`

Competency question: What service is indicated by a service sign?

Method: DL Query

Protégé option: Subclasses

Translation:

`RoadSign and (indicatesService some Service)`

Competency question: Which additional panel modifies a main road sign?

Method: DL Query

Protégé option: Subclasses

Translation:

`RoadSign and (modifiesSign some RoadSign)`

Competency question: Which main sign is modified by an additional panel?

Method: DL Query

Protégé option: Subclasses

Translation:

`RoadSign and (isModifiedByPanel some AdditionalPanel)`

Note: This requires restrictions using `isModifiedByPanel`; otherwise use the inverse property axiom in SPARQL.

Competency question: Can the ontology distinguish between signs expressing prohibitions, obligations, restrictions,

hazards, directions, services, and priority rules?

Method: DL Query group

Protégé option: Subclasses

Translation:

`RoadSign and (expressesProhibition some Prohibition)`

`RoadSign and (expressesObligation some Obligation)`

`RoadSign and (expressesRestriction some Restriction)`

`RoadSign and (warnsAbout some RoadHazard)`

`RoadSign and (indicatesDirection some Direction)`

`RoadSign and (indicatesService some Service)`

`RoadSign and (indicatesPriorityRule some PriorityRule)`

Note: Run the group of queries and compare the returned classes.

4. CQs relevant to the particular dataset: MTDS

Competency question: Which MTDS classes are represented in the ontology?

Method: DL Query

Protégé option: Subclasses

Translation:

`MTDSDatasetSignClass`

Competency question: Which MTDS classes are subclasses of existing road-sign categories?

Method: DL Query

Protégé option: Subclasses

Translation:

`MTDSDatasetSignClass and RoadSign`

Competency question: Which MTDS classes are equivalent to Vienna Convention road-sign classes?

Method: SPARQL

Translation:

```
SELECT ?class ?equivalentClass
WHERE {
  ?class owl:equivalentClass ?equivalentClass .
  FILTER(CONTAINS(STR(?class), "MTDS") || CONTAINS(STR(?equivalentClass), "MTDS"))
}
ORDER BY ?class
```

Competency question: Are all MTDS classes connected to the RoadSign hierarchy?

Method: DL Query

Protégé option: Subclasses

Translation:

MTDSDatasetSignClass and RoadSign

Competency question: Which MTDS classes belong to warning signs?

Method: DL Query

Protégé option: Subclasses

Translation:

MTDSWarningSignClass

Competency question: Which MTDS classes belong to regulatory signs?

Method: DL Query

Protégé option: Subclasses

Translation:

MTDSRegulatorySignClass

Competency question: Which MTDS classes belong to mandatory signs?

Method: DL Query

Protégé option: Subclasses

Translation:

MTDSMandatoryRegulatoryClass

Competency question: Which MTDS classes belong to prohibitory signs?

Method: DL Query

Protégé option: Subclasses

Translation:

```
MTDSProhibitoryRegulatoryClass
```

Competency question: Which MTDS classes belong to direction or indication signs?

Method: DL Query

Protégé option: Subclasses

Translation:

```
MTDSDirectionIndicationClass
```

Competency question: Are there MTDS classes that do not have an exact Vienna Convention equivalent?

Method: SPARQL

Translation:

```
SELECT ?mtdsClass
WHERE {
  ?mtdsClass rdf:type owl:Class .
  FILTER(CONTAINS(STR(?mtdsClass), "MTDS"))
  FILTER NOT EXISTS { ?mtdsClass owl:equivalentClass ?equiv . }
  FILTER NOT EXISTS { ?equiv owl:equivalentClass ?mtdsClass . }
}
ORDER BY ?mtdsClass
```

Note: Finds MTDS classes without explicit equivalentClass mappings.

Competency question: What MTDS dataset label is associated with a class?

Method: SPARQL

Translation:

```
SELECT ?class ?label
WHERE {
  ?class ?p ?label .
  FILTER(CONTAINS(STR(?p), "hasMTDSClassLabel"))
```

```
}
```

```
ORDER BY ?class
```

Competency question: What image filename contains a given MTDS annotation?

Method: SPARQL

Translation:

```
SELECT ?annotation ?image ?filename
WHERE {
  OPTIONAL { ?annotation ?rel ?image . FILTER(CONTAINS(STR(?rel), "Image")) }
  ?image ?p ?filename .
  FILTER(CONTAINS(STR(?p), "hasImageFilename"))
}
```

Competency question: What are the width and height of an MTDS image?

Method: SPARQL

Translation:

```
SELECT ?image ?width ?height
WHERE {
  OPTIONAL { ?image ?wProp ?width . FILTER(CONTAINS(STR(?wProp), "hasImageWidth")) }
  OPTIONAL { ?image ?hProp ?height . FILTER(CONTAINS(STR(?hProp), "hasImageHeight")) }
}
```

Note: Requires image individuals with data values. For schema validation, query the domains/ranges of hasImageWidth and

hasImageHeight.

Competency question: What dataset split does an MTDS image belong to?

Method: SPARQL

Translation:

```
SELECT ?image ?split
WHERE {
  ?image ?p ?split .
  FILTER(CONTAINS(STR(?p), "hasDatasetSplit"))
}
```

```
}
```

Competency question: What are the bounding-box coordinates of an MTDS annotation?

Method: SPARQL

Translation:

```
SELECT ?annotation ?xmin ?ymin ?xmax ?ymax

WHERE {

  OPTIONAL { ?annotation ?xminProp ?xmin . FILTER(CONTAINS(STR(?xminProp),
"hasBoundingBoxXMin")) }

  OPTIONAL { ?annotation ?yminProp ?ymin . FILTER(CONTAINS(STR(?yminProp),
"hasBoundingBoxYMin")) }

  OPTIONAL { ?annotation ?xmaxProp ?xmax . FILTER(CONTAINS(STR(?xmaxProp),
"hasBoundingBoxXMax")) }

  OPTIONAL { ?annotation ?ymaxProp ?ymax . FILTER(CONTAINS(STR(?ymaxProp),
"hasBoundingBoxYMax")) }

}
```

Competency question: Are MTDS metadata properties clearly distinguished from road-sign semantic properties?

Method: SPARQL

Translation:

```
SELECT ?property ?domain ?range

WHERE {

  ?property rdf:type owl:DatatypeProperty .

  OPTIONAL { ?property rdfs:domain ?domain . }

  OPTIONAL { ?property rdfs:range ?range . }

  FILTER(

    CONTAINS(STR(?property), "MTDS") ||

    CONTAINS(STR(?property), "Image") ||

    CONTAINS(STR(?property), "Annotation") ||

    CONTAINS(STR(?property), "BoundingBox")

  )

}

ORDER BY ?property
```

